

Real-Time Speech & Voice Signal Processing for Browser-Based Coaching Tools

Executive Summary

Based on current research and industry benchmarks, real-time browser-based speech analysis has hard technical and ethical boundaries. Filler word detection and pause analysis are feasible for live coaching with known accuracy tradeoffs. Voice tone and emotion classification are NOT recommended for real-time coaching feedback—they lack sufficient accuracy, generalize poorly across populations, and carry ethical risks.

1. Filler Word Detection ("Um," "Uh," "Like," "You Know")

Current State of the Art

Benchmark Results (PodcastFillers Dataset, 2022):

- AVC-FillerNet (ASR-based): 71-78% F1 score | Requires full transcription
- VC-FillerNet (VAD-only, no ASR): 64.8% F1 score | ~50-100ms latency
- Gentle baseline: 64-70% F1 score
- "Um" vs "Uh" distinction: 77-78% F1 ("um") vs 64.8% ("uh")—"um" is significantly easier due to longer duration

Browser-Based Implementation Options

Option 1: Lightweight KWS Approach (Real-Time Safe)

How it works:

- Voice Activity Detection (VAD) identifies potential filler locations
- Lightweight neural classifier (TC-ResNet8, ~100k parameters) discriminates fillers from speech
- Uses wav2vec embeddings (pre-computed, small footprint)
- Runs entirely on client via WebAudio API + ONNX.js

Pros:

- 50-100ms latency (suitable for live feedback)
- Works offline in browser
- No server dependency

Cons:

- 65% F1 score (5-10 percentage points lower than ASR-based)
- Higher false positive rate; requires user tolerance for occasional misclassifications
- Language-specific; English models only currently published

Browser Support: Chrome, Edge, Firefox (with ONNX.js polyfill)

Option 2: Streaming ASR + Post-Processing (Post-Session Only ✖)

How it works:

- Send audio to Deepgram/AssemblyAI streaming API
- ASR generates transcript; identify gaps where VAD activates but ASR produces silence
- Classifier determines if gap is filler or natural pause

Pros:

- 71-78% F1 score (state-of-the-art)
- Highly accurate; separates "um" vs "uh"
- Can combine with sentiment/diarization

Cons:

- 300-500ms latency to first detection (too slow for real-time coaching cues)
- Requires external API calls (~\$0.0077/min Deepgram streaming)
- Can't provide live feedback during session

Recommended use: Post-session analytics dashboard; coaching review materials

Accuracy Considerations for Coaching Context

Key limitation: False positive rate on technical speech (e.g., presenter saying "uhhh, let me explain the architecture").

- Fillers ≠ poor communication; can reflect thoughtful processing

- Risk of overuse: coaching tool that flags every filler may create speech anxiety rather than improvement
 - Recommendation: Use as input to larger coaching logic, not as standalone metric
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2. Pause Detection & Classification

Pause Detection (Simple VAD)

Benchmark: 92% precision/recall on clean audio; 70-80% in background noise

Browser Implementation (Real-Time Safe 

javascript

```
// Web Audio AnalyserNode approach
const analyser = audioContext.createAnalyser();
const threshold = 19; // dB relative to peak amplitude
// Frame-by-frame classification: speech vs. silence
// Latency: 20-50ms per frame
```

Open-source libraries:

- Silero VAD (ONNX, ~3MB)—10-20ms latency, 90%+ accuracy on clean audio
- Picovoice Porcupine—Commercial tier available; <20ms latency
- Simple amplitude thresholding—Requires speaker calibration; fails in noise

Pause Classification: Cognitive vs. Hesitation-Related

The Challenge:

Research shows pause length and post-pause speech behavior (rate recovery, pitch reset, breath detection) can distinguish:

- Cognitive pauses (thinking time): longer, followed by normal resumption
- Hesitation pauses (uncertainty): variable length, followed by speech rate decrease, pitch elevation

Benchmark & Limitations:

Feature Set	Accuracy	Dataset	Notes
Pause length alone	~60%	Mild cognitive impairment studies	Overlapping distributions; high variance
Pause + post-pause rate	~70-80 %	Clinical speech data	Requires end-to-end utterance analysis
Pause + pitch reset + rate	~75-85 %	Limited validation	Few published studies on coaching contexts

Known Failure Modes:

1. Speaker baseline variance: Individuals naturally have different pause patterns; no universal threshold
2. Noise sensitivity: Background noise breaks VAD; false pauses detected
3. Linguistic confounds: Language structure (e.g., conjunction "and" naturally followed by pause) mimics hesitation
4. Coaching-specific problem: Thoughtful leaders pause frequently without hesitation

Browser Implementation (Delayed Feedback Only ⚠️):

- Requires buffering until pause ends + analyzing post-pause segment
- Minimum 500ms delay from pause onset to classification
- Post-event analysis only (unsuitable for real-time coaching cues)

Recommendation: Use for post-session review; risky for real-time feedback (may interrupt natural thinking patterns).

3. Prosody & Cadence Analysis

Pitch/Frequency Extraction

Browser-Native Approach (Real-Time Safe 

```
javascript

// Autocorrelation via Web Audio API
const buffer = new Float32Array(analyser.fftSize);
analyser.getFloatTimeDomainData(buffer);
// Autocorrelation: find period where signal correlates with itself
// F0 = sampleRate / period
```

Characteristics:

- Latency: 20-50ms (one FFT window)
- Accuracy: ±5% Hz on sustained tones; worse on speech (±10-15%)
- Limitation: Octave errors (detecting 2×F0 instead of F0); mitigated by smaller FFT windows
- Cost: Free; included in Web Audio API

Prosody Metrics for Coaching

Measurable in real-time:

Metric	How Measured	Benchmark	Limitation
Pitch range (min-max)	F0 extraction over utterance	No standard (speaker-specific)	Confounded by gender, age, vocal physiology
Pitch excursion	Semitone range	Research: Engaging speakers = 6-8 semitone range; Monotone = <3	Highly variable; genre-dependent (lecturing vs. conversing)

Speech rate	Words-per-minute; syllables/sec	~140-150 WPM typical; confident = slightly faster; anxious = slower	Requires noise-robust VAD; fails in background noise
Rhythm smoothness (jitter)	Micro-F0 variation (10-20Hz fluctuation)	No clinical benchmark for coaching	Jitter ↑ = vocal fatigue/tension; but also ↑ with emotion, not just anxiety
Energy envelope variance	Loudness fluctuation analysis	No standard metric	Confounded with room acoustics, microphone distance

Commercial Solutions for Prosody

Reality: No commercial API currently offers real-time prosody analysis designed for coaching.

- Deepgram, AssemblyAI: Transcription-focused; no prosody extraction
- Research tools (PRAAT, OpenSMILE): Offline analysis; requires audio files
- Custom ML: Requires training on domain-specific coaching data (rare)

Why it matters for your platform:

Prosody is one of the most reliable markers of genuine vs. coached speech. If you can quantify pitch range, rate, and energy variance in real-time, you have a unique coaching signal (with caveats below).

Recommendation for MetaMind

Feasible real-time metrics:

1. Speech rate (syllables/sec) — Robust; simple to compute
2. Pitch range (semitones) — Noisy but real-time detectable
3. Pause frequency & duration — Simple VAD

NOT ready for real-time:

1. Rhythm smoothness / jitter (insufficient validation; easily confounded)

2. Formant analysis (too complex for browser; requires spectral decomposition)
 3. Vocal fry / creaky voice (hard to detect in noisy environments)
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4. Voice Tone Classification: Critical Limitations

What Can Be Technically Detected (Without Emotional Attribution)

Acoustic features that are technically measurable:

- Pitch elevation (↑ = arousal, anxiety, or excitement)
- Loudness decrease (↓ = uncertainty or resignation; can also = politeness)
- Speech rate decrease (↓ = cognitive load, hesitation, or deliberate emphasis)
- Breathiness (↑ = vocal fatigue, anxiety, or intimacy)

What CANNOT be reliably inferred:

- Emotional state (angry, happy, sad, anxious)
- Speaker confidence level
- Speaker intent or truthfulness
- Speaker authenticity vs. acting

Why Tone Classification Fails in Coaching Contexts

Problem 1: Bidirectional Causality

A coach might elevate pitch and speak faster while COACHING (intentional, confident behavior) vs. while being coached (anxious, uncertain behavior). The acoustic features are identical—only context reveals meaning.

Problem 2: Cultural & Demographic Bias

Accuracy drop-off:

- Models trained on Western English speakers: 70-85% accuracy
- Same models on non-native English speakers: 45-60% accuracy
- Models fail on neurodivergent speakers (autism spectrum, ADHD) who have atypical prosody
- Gender bias: Female speakers perceived as "less confident" due to higher pitch baseline

Problem 3: Confidence Scores Are Unreliable

Research shows ML models are overconfident even when wrong. A classifier reporting 95% confidence that a speaker is "anxious" may be 40% accurate on real data.

Problem 4: Context Collapse

Sarcasm, irony, playfulness, and technical concentration all produce acoustic patterns similar to anxiety or low confidence. Without transcription + semantic understanding, misclassification is guaranteed.

Published Benchmarks (Emotion Recognition)

Task	Accuracy	Notes
Emotion detection (within-speaker)	70-85%	Speaker-specific models; limited transferability
Emotion detection (across-speaker)	45-65%	Dramatic drop; severe generalization failure
Non-native English speakers	35-50%	Significant bias
Confidence classification	~60-75%	Better than full emotion, but still unreliable

Source: NIH studies on sentiment analysis in mental health chatbots; ACM CHI 2024 emotion recognition ethics reviews

Ethical Concerns Specific to Coaching

⚠️ DO NOT use voice tone classification for:

- Real-time feedback during coaching ("I'm detecting anxiety in your voice...")

- Performance metrics ("Your confidence score is 62%...")
- Hiring or evaluation decisions based on speech analysis alone

Risks:

1. Discrimination: Misclassifying non-native speakers or neurodivergent individuals as less confident
2. Psychological harm: Creating speech anxiety in coachees who are told their voice indicates low confidence
3. Misjudgment: Flagging thoughtful pauses or vocal variety as "uncertainty"
4. Manipulation: Coaching tool could reinforce vocal patterns that are performative rather than authentic

Recommendation for MetaMind

What to avoid:

- Real-time emotion/tone classification (insufficient accuracy)
- Confidence scoring based solely on prosody (bidirectional causality problem)
- Cross-demographic generalization (severe bias risk)

What's feasible with caveats:

- Post-session analytics showing pitch range, rate, and pause patterns as descriptive metrics (not predictive)
- Coaching prompts like: "You paused for 2.3 seconds here. What were you thinking?"—coaching reflection, not diagnosis
- Speech rate tracking over multiple sessions (trend analysis is more reliable than single-utterance classification)

5. Real-Time vs. Post-Session: Decision Framework

Deployable in Real-Time (Green Light)

Technique	Latency	Accuracy	Cost	Notes
Silence detection (VAD)	10-20ms	90%+ clean; 70% noise	Free	Use for turn-taking signals

Speech rate	100-300 ms	High (frame-counting)	Free	Descriptive; no interpretation
Pitch range	50ms	±10% Hz	Free	Noisy; use trending, not single value
Basic pause detection	50ms	92% clean	Free/Low-cost (Siler)	Use for coaching interventions

Requires Delayed Feedback (Caution ⚠)

Technique	Delay	Accuracy	Cost	Notes
Filler detection (VAD-only KWS)	50-100ms	65% F1	Open-source	Post-phrase feedback OK; not live
Pause classification	500ms+	70-80%	Custom ML	Must wait for pause to end + analysis window
Post-pause rate change	1-2s	75%	Custom	Requires utterance boundary detection
Rhythm analysis	1-3s	~60%	Research-grade	Clinical validation lacking

DO NOT Deploy Real-Time (Red Light ❌)

Technique	Why Not	Safer Alternative
Emotion/tone classification	Accuracy <65% cross-speaker; severe bias; bidirectional confound	Post-session trends; transcription-based sentiment
Confidence scoring (audio-only)	Can't distinguish authentic vs. performed speech	Coaching reflection: "Tell me what you were thinking when..."
Stress/anxiety detection	Identical acoustic features to excitement, careful thinking, engagement	Context-aware analysis (what was being discussed?)
Filler detection (real-time feedback)	May increase speech anxiety; disrupts natural thinking	Post-session review or trending over time
Vocal fry / creaky voice	Hard to detect reliably; confounded with emotion, fatigue, genre	Manual observation by human coach

6. Latency Breakdown: The 300ms Rule

For real-time coaching interactions, aim for <300ms latency from audio to actionable insight.

Latency budget breakdown (for reference):

text

Audio capture: 10-50ms

Network upload: 20-100ms

ASR processing: 100-200ms (streaming)

Analysis: 50-100ms (prosody/pause)

Network download: 20-100ms
UI rendering: 50-100ms

Total: 250-650ms (target <300ms for snappy UX)

For coaching feedback to feel natural:

- Turn-taking cues (pause detection): <100ms acceptable
- Session metrics (speech rate): <500ms; real-time display optional
- Coaching prompts ("Tell me more"): <2s acceptable

7. Open-Source vs. Commercial APIs: Cost-Benefit

Open-Source Options (Self-Hosted)

Library	Accuracy	Latency	Cost	Best For
Silero VAD	90% clean; 70% noise	10-20ms	Free	Pause/silence detection
wav2vec (Meta)	85-90% WER	CPU-dependent; ~500ms	Free + compute	Offline ASR; not browser
Whisper (OpenAI)	90-95% WER	Slow on CPU; <300ms with GPU	Free	Offline transcription; post-session
SpeechBrain (PyTorch)	75-90% WER	Requires training/tuning	Free	Customization; research

Commercial APIs (Cloud)

API	Accuracy	Latency	Cost	Best For
Deepgram Nova-3	90-95% WER; diarization	<300ms TTFB	\$0.0077/min streaming	Production ASR; real-time transcription
AssemblyAI Universal-Stream ing	90-95% WER; sentiment	~270ms TTFB	\$0.000833/min	Real-time transcription + sentiment (limited)
Google Speech-to-Text	85-95% WER (varies)	500-2000ms	Free tier + pay-as-you-go	Web Speech API fallback; not recommended for production
AWS Transcribe	85-92% WER	1-5s (batch); 1-2s (streaming)	\$0.0001/min	Batch processing; not real-time

Recommendation for MetaMind

For real-time coaching:

1. Use Deepgram or AssemblyAI streaming for transcription backbone (99.9% SLA)
2. Layer browser-based VAD + speech rate for real-time session insights
3. Store raw audio + transcript for post-session analysis
4. Add open-source filler detection (KWS) as optional, lower-accuracy real-time layer

Cost estimate (1000 coaches, 1 hour/week coaching each):

- Deepgram: $\sim 4000 \text{ min/week} \times \$0.0077 = \$30.80/\text{week}$ or $\sim \$1600/\text{year}$
- AssemblyAI: $\sim 4000 \text{ min/week} \times \$0.000833 = \$3.33/\text{week}$ or $\sim \$173/\text{year}$
- Browser-based VAD/pitch: \$0 (after initial implementation)

8. Ethical Guardrails for Implementation

What to Disclose to Users

If deploying real-time speech analysis:

-  "We're analyzing your speech rate and pause patterns to provide coaching feedback."
-  "These metrics are descriptive only—not predictive of confidence or emotion."
-  "You can see how your speech evolves over time in the session review."
-  Avoid: "Your anxiety level is 72%"
-  Avoid: "Your confidence dropped 15% in this sentence"
-  Avoid: "You sound uncertain" (without contextual coaching)

Consent & Bias Testing

Before deploying voice tone or emotion classification:

1. Test across demographic groups: gender, non-native English speakers, age
2. Measure accuracy drop-off: If accuracy falls >20 percentage points on any group, don't deploy
3. Disclose limitations: "This feature may be less accurate for non-native English speakers"
4. Allow opt-out: Let users disable real-time voice analysis without losing session access

Coaching Context Check

Ask: "Is this metric actually helpful for coaching, or does it create anxiety?"

Example: Flagging "um" in real-time may cause coachees to suppress natural speech patterns rather than improve underlying communication skills. Post-session review of filler frequency allows coaching reflection without live disruption.

9. Implementation Checklist for Your Platform

Phase 1: MVP (Deployable Now)

- Transcription backbone: Deepgram or AssemblyAI streaming API
- Session metrics: Speech rate (syllables/sec), pause frequency, duration
- VAD for turn-taking: Silero VAD ONNX in browser
- Post-session dashboard: Display speech rate trends, pause patterns over time
- Coaching prompts: Rule-based ("You paused 5 times in that response. What were you thinking?")

Phase 2: Enhanced Analytics (3-6 Months)

- Lightweight filler detection: VAD + KWS classifier (64% F1) as optional layer

- Pitch range tracking: Real-time F0 extraction; trending over sessions
- Silence pattern analysis: Detection of natural pauses vs. disfluencies
- Transcript-based sentiment: Analyze coachee's own words (more reliable than voice tone)

Phase 3: Research-Grade (Future, Validated on Coaching Data)

- Pause classification: Cognitive vs. hesitation (only after clinical validation on coaching population)
- Individual baseline modeling: Personalized prosody patterns (reduce bias from population-level models)
- Multimodal coaching: Combine speech + transcript + video (if available) for richer insights

Phase 4: DO NOT DO

- ❌ Real-time emotion/confidence classification from voice alone
 - ❌ Cross-system deployment of pre-trained sentiment models without in-domain validation
 - ❌ Confidence scoring that lacks human-in-the-loop review
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10. Research Gaps & Unknowns

For Coaching Specifically

1. Filler detection in coaching contexts: Published benchmarks are on podcasts/interviews. How does accuracy change when participants are being coached (higher cognitive load)?
2. Pause classification patterns in coaching: Do cognitive vs. hesitation pauses have different profiles during coaching vs. other speech tasks?
3. Prosody feedback effect: Does real-time feedback on pitch/rate improve communication skills or increase anxiety? No RCT data available.
4. Coachee preferences: What speech analysis metrics do coachees find helpful vs. distracting? User research needed.

Open Questions

- Can post-pause speech features (rate, pitch reset) reliably distinguish confidence levels in coached populations?

- What is the optimal latency for coaching feedback? (Current literature suggests <2s, but no coaching-specific UX studies)
- How do filler words actually relate to communication effectiveness? (Assumption: more fillers = worse. Reality: may vary by domain, context, audience.)

Summary: Safe Deployment Boundaries

Metric	Real-Time ?	Accuracy	Cost	Notes
Silence/pause detection	✅ Yes	90%	Free	Use for turn-taking
Speech rate	✅ Yes	High	Free	Descriptive, no interpretation
Pitch range (trending)	✅ Yes	±10%	Free	Use multi-session trends, not single values
Filler detection (KWS)	⚠️ Delayed	65% F1	Open-source	Post-utterance feedback only
Pause classification	⚠️ Delayed	70-80%	Custom	Requires end-to-pause analysis
Voice tone / emotion	❌ No	45-65%	API	Biased; ethically risky; insufficient accuracy

Confidence scoring (audio)	✗ No	50-70%	Custom	Bidirectional confound; unreliable
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Fundamental Rule: If it requires interpreting speaker intent, emotion, or confidence from audio alone, you need human coaching judgment. Metrics should be descriptive (what happened) not prescriptive (what it means).

AssemblyAI. "The 300ms Rule: Why Latency Makes or Breaks Voice AI Applications." (2025). Describes the perceptual threshold for real-time voice interactions.

Zhu, G., Caceres, J., & Salamon, J. (2022). "Filler Word Detection and Classification: A Dataset and Benchmark." Interspeech. PodcastFillers dataset; AVC-FillerNet achieves 78.4% F1 on "um," 64.8% on "uh."

Sakhnov, K., et al. (2021). Voice Activity Detection methods in automatic speech recognition. NIH study on pause extraction in Alzheimer's disease detection; VAD accuracy 92% precision/recall on clean audio.

University of Miami Miller School of Medicine. (2024). "Post-Pause Speech Patterns Help Detect Mild Cognitive Impairment." Demonstrates post-pause speech behavior (rate, pitch reset) correlates with cognitive state.

Chatziagapi, et al. (2022). "Audio and ASR-based Filled Pause Detection." ACII. Shows transfer learning from speaking rate and emotion classification improves filler detection by ~1%.

Deepgram & Cresta. (2025). "Real-Time Voice Agent Latency." Analysis of semantic turn detection; notes that context-based (not audio-only) confidence scoring is necessary for coached interactions.

Gaslighting Check. (2025). "7 Ways AI Reads Your Emotions Through Voice." Meta-analysis showing 20-30% accuracy drop-off for non-native English speakers and significant gender bias in pitch-based confidence scoring.

NIH. (2024). "Ethical Aspects of Integrating Sentiment and Emotion Analysis in Digital Therapeutics." PMC11602467. Documents risks of low-quality emotion analysis in mental health contexts; identifies bias and context-collapse as primary failure modes.

ACM CHI 2024. (2024). "Collecting and Implementing Ethical Guidelines for Emotion AI." Highlights risks of manipulation and autonomy erosion when emotion detection is used to tailor coaching feedback.

Gladia & Cresta. (2025). "Measuring Latency in Speech-to-Text." TTFB $\leq 100\text{ms}$ perceived as "instantaneous"; final latency $\leq 700\text{ms}$ acceptable for real-time dialogue.

Document prepared for MetaMind behavioral diagnostic platform development. For coaching-specific validation of these techniques, user research and pilot studies with your target coaching population are recommended before full deployment.

Research validation methodologies, bias risks, and claim-safe practices for speech and voice analysis systems used in coaching, training, or productivity tools (non-clinical).

Focus areas:

1. Validation standards

- How accuracy is typically validated for:
 - filler word detection
 - speech rate and pause metrics
 - pitch and prosody features
- Ground-truth labeling methods
- Inter-rater reliability benchmarks
- What accuracy thresholds are considered "good enough" for behavioral feedback (not diagnosis)

2. Bias and generalization

- Accuracy degradation across:
 - non-native English speakers
 - gender
 - age
 - neurodivergent speech patterns
- Published evidence of bias in prosody-, pitch-, or rate-based models

- Mitigation strategies used in production systems (personal baselining, trend-based feedback, opt-outs)

3. Claim framing and uncertainty

- How reputable products and papers frame uncertainty
- Difference between descriptive metrics vs interpretive or predictive claims
- Language patterns used to avoid emotional or psychological attribution

4. Regulatory and ethical guidance (non-clinical)

- Relevant guidance from ACM, IEEE, NIH, EU AI Act (if applicable)
- What types of claims trigger higher legal or ethical scrutiny
- Best practices for disclosures and user consent in voice analysis tools

Output format:

- Validation checklist (what must be true before a metric is shown to users)
- Bias risk table (risk, affected groups, mitigation)
- "Safe to claim" vs "Do not claim" language examples
- Citations to primary sources

Excellent research. Now I have comprehensive data on validation standards, bias mitigation, regulatory guidance, and claim framing. Let me compile this into a detailed document with actionable checklists and examples.